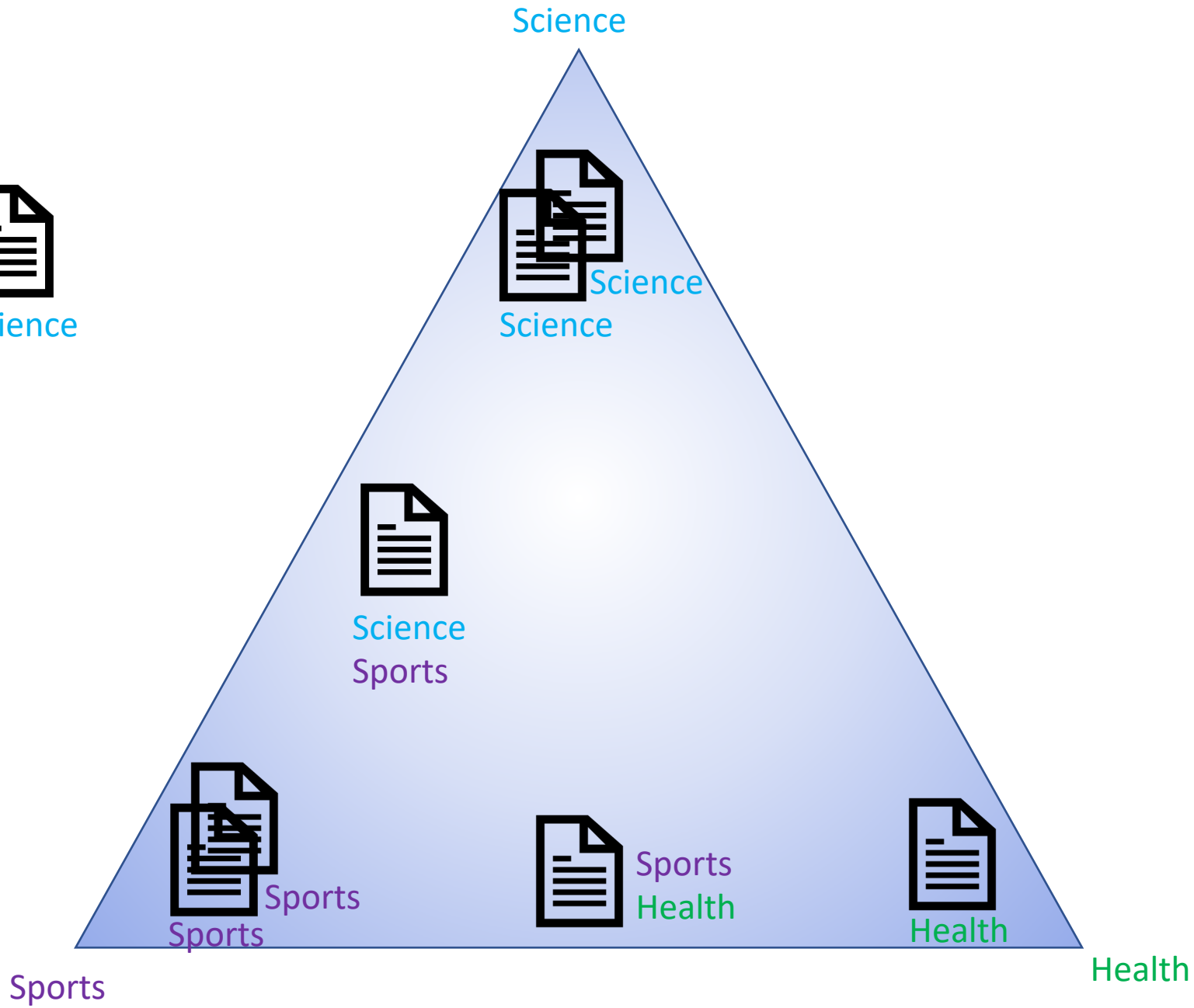
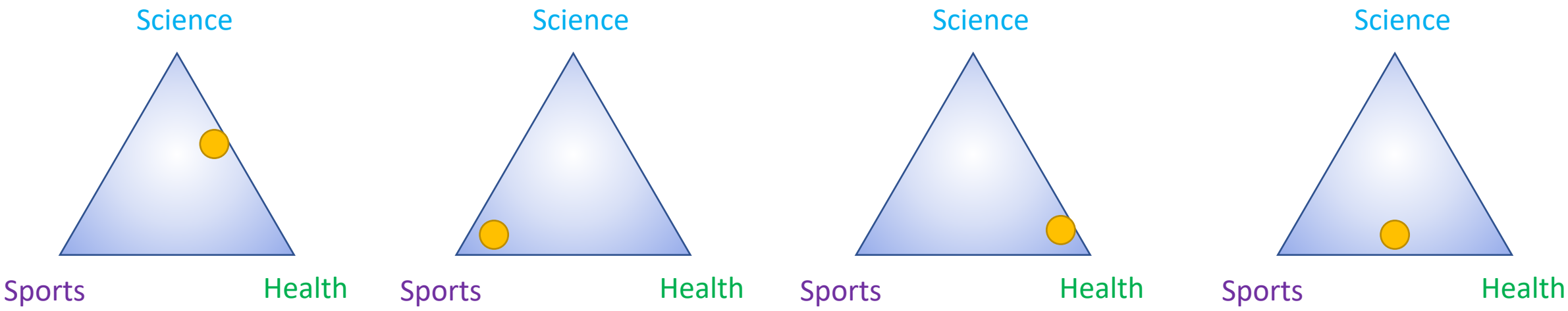
Latent Dirichlet Allocation
LDA

The problem

Science
Sports
Health
Science
Sports
Sports
Health
Science
Sports

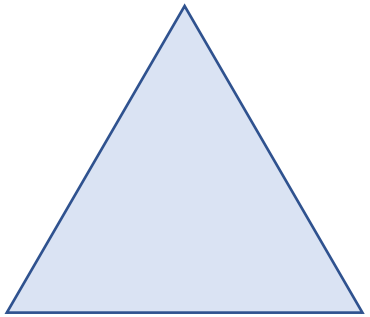

Science
Science
Science
Science
Sports
Sports
Sports
Sports
Health
Health
Health

Doc 1	Doc 2	Doc 3	Doc 4
World	Football	World	World
Covid-19	Football	World	Football
Scientist	Football	Outbreak	Covid-19
Covid-19	World	Covid-19	Football
Scientist	World	Covid-19	World

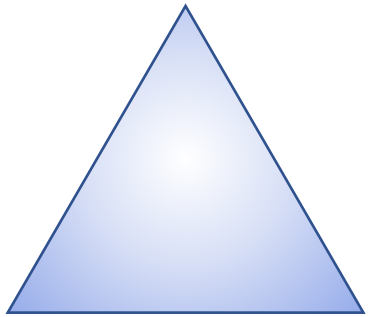


Dirichlet distribution

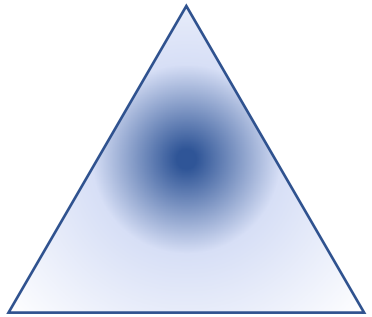
$$f\left(x_1, \ldots, x_K; \alpha_1, \ldots, \alpha_K\right) = \frac{1}{\mathrm{B}(\boldsymbol{\alpha})} \prod_{i=1}^{K} x_i^{\alpha_i - 1}$$



$\propto = 1$

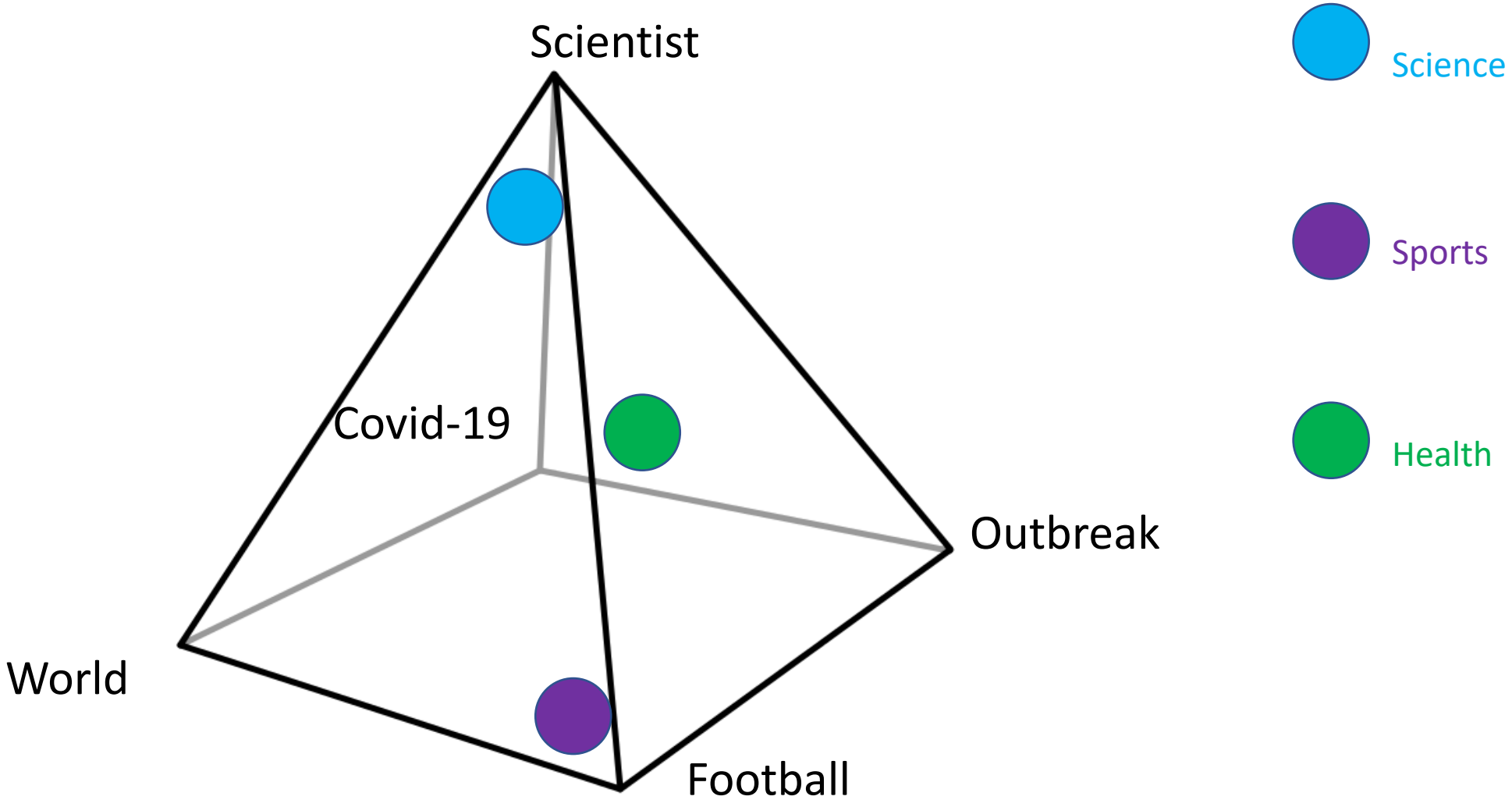


$\propto < 1$

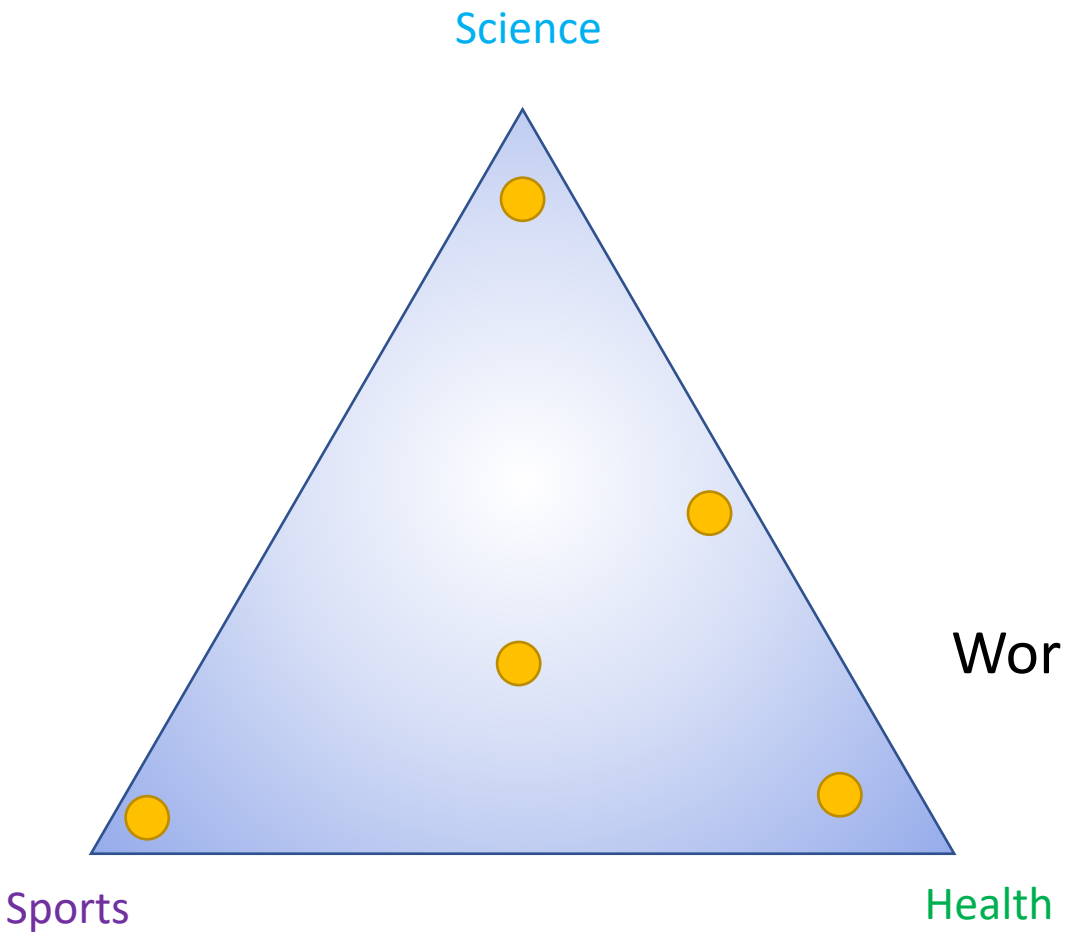


$\propto > 1$

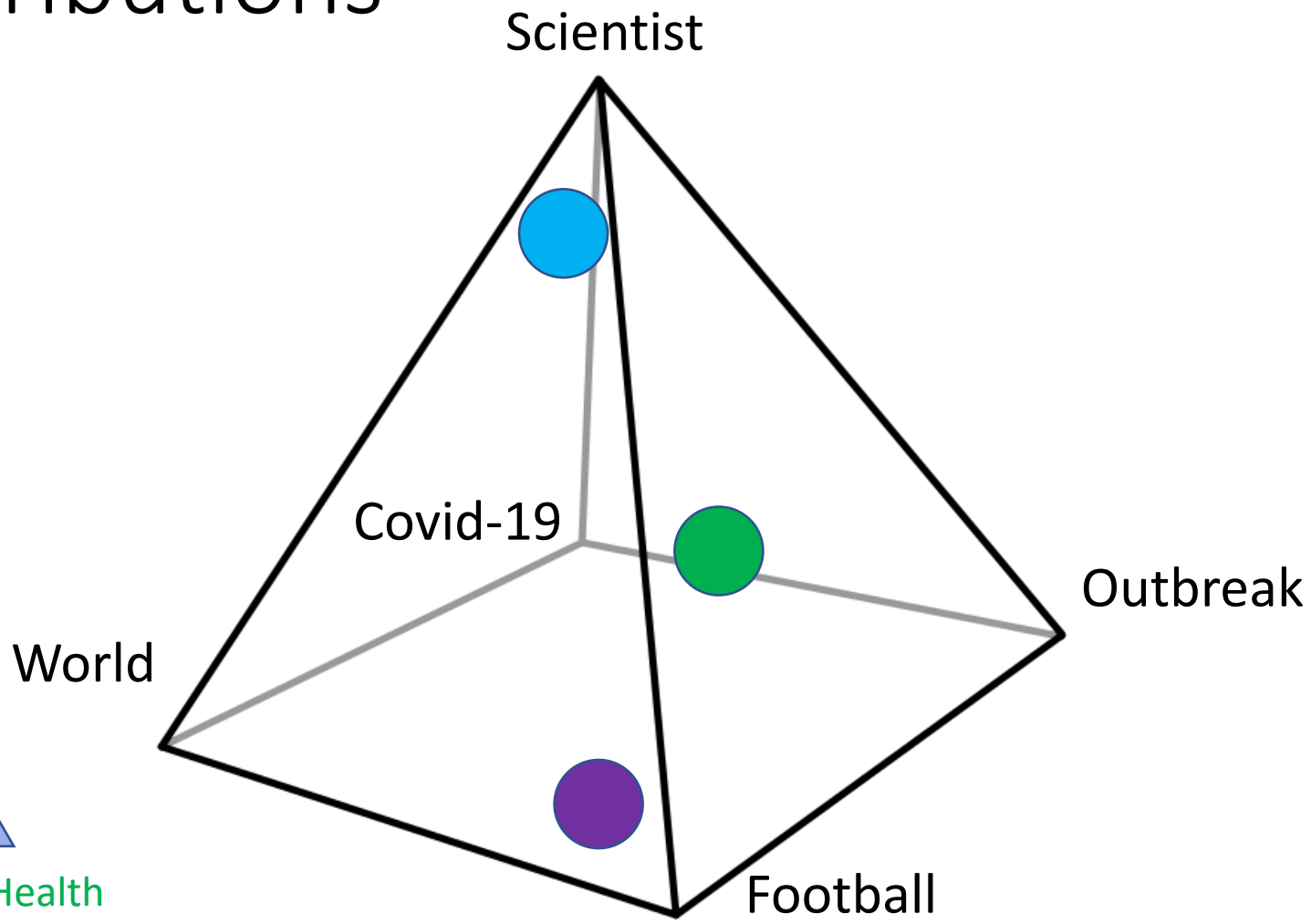
Where we put the topics



Two Dirichlet distributions

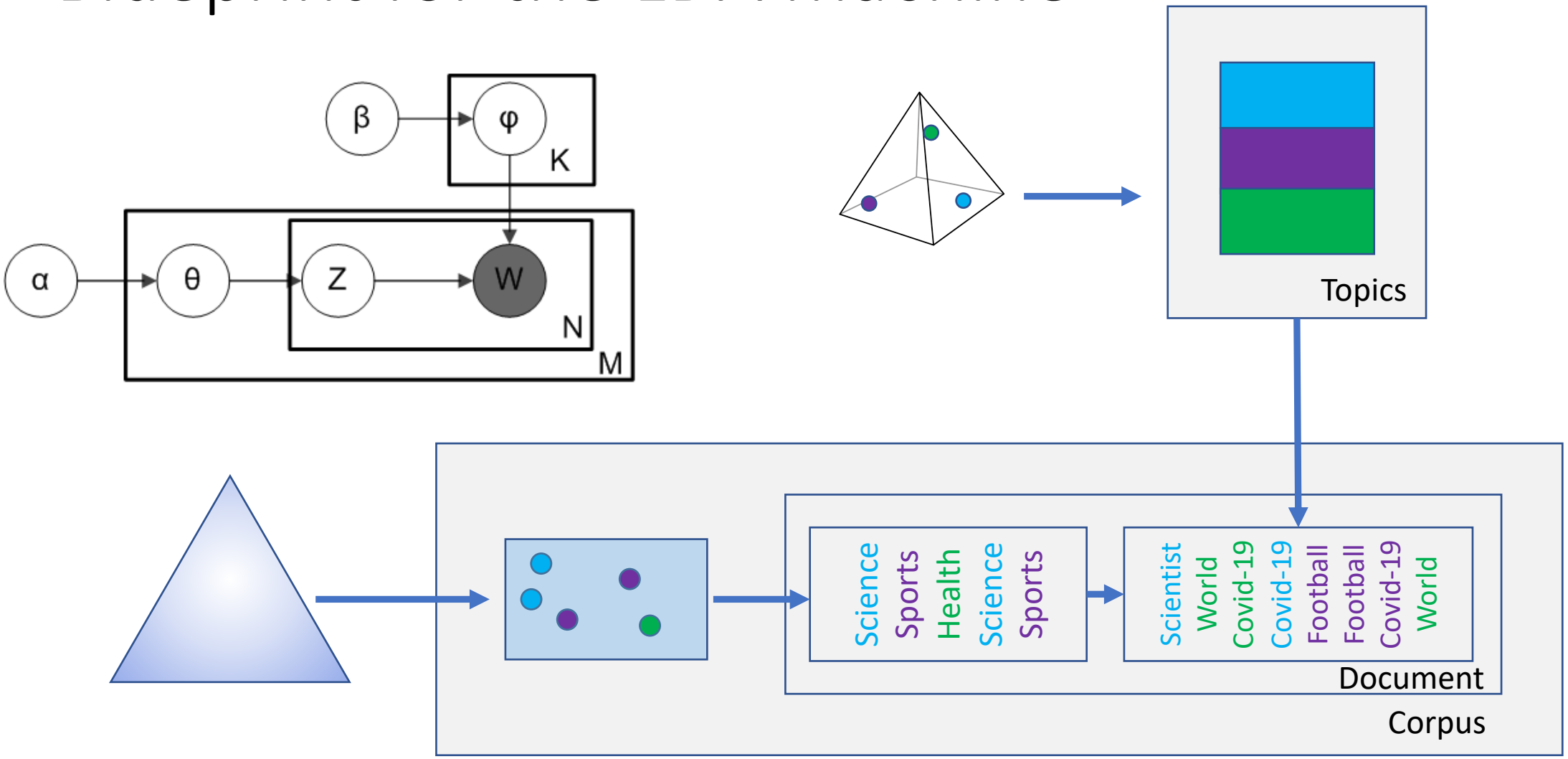


Documents-Topics



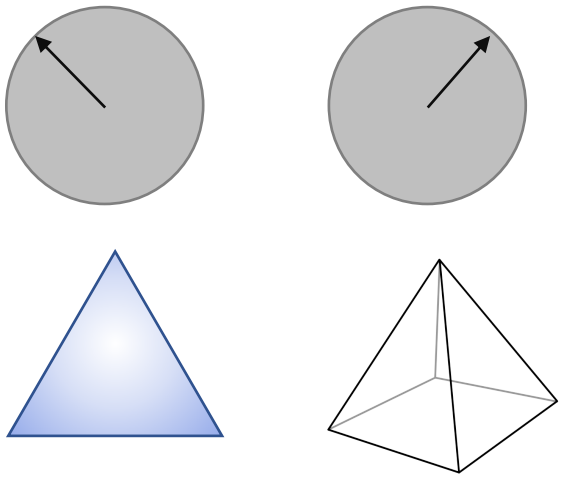
Topics-Words

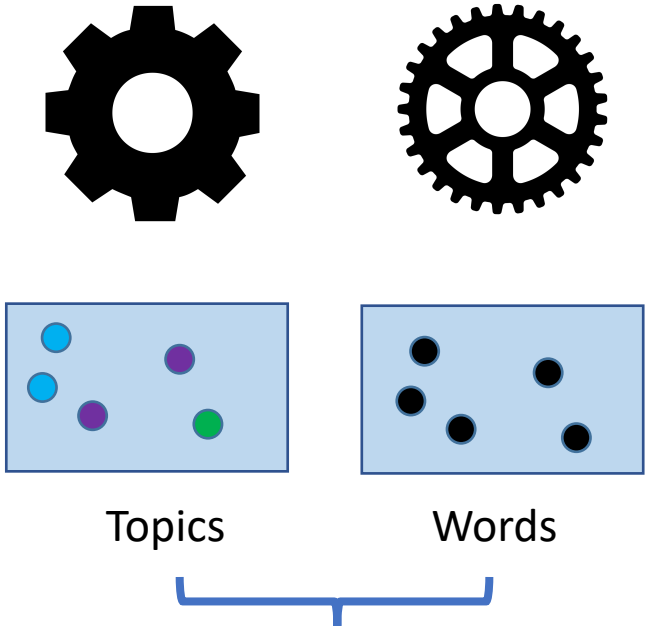
Blueprint for the LDA machine



Probability of a document

$$P(\boldsymbol{W}, \boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{\varphi}; \alpha, \beta) = \prod_{i=1}^{K} P(\varphi_i; \beta) \prod_{j=1}^{M} P(\theta_j; \alpha) \prod_{t=1}^{N} P(Z_{j,t} \mid \theta_j) P(W_{j,t} \mid \varphi_{Z_{j,t}})$$





Dirichlet distributions

Multinomial distributions

Gibbs Sampling

*Probability of word **w** under topic **t***

$$P(z_i = j \mid \mathbf{z}_{-i}, w_i, d_i, \cdot) \propto \frac{C^{WT}_{w_i j} + \beta}{\sum_{w=1}^{W} C^{WT}_{wj} + W\beta} \frac{C^{DT}_{d_i j} + \alpha}{\sum_{t=1}^{T} C^{DT}_{d_i t} + T\alpha}$$

Probability of topic for a word in a document

*Probability of topic **t** in document **d***

Run Gibbs Sampling

- Start: assign each word token to a random topic
- $\mathbf{C}^{WT}$ = Count number of times a word token ***wi*** was assigned to a topic ***j***
- $\mathbf{C}^{DT}$ = Count number of times a topic ***j*** was already assigned to some word token in doc ***di***
- First Iteration:
  - For each word token, the count matrices $\mathbf{C}^{WT}$ and $\mathbf{C}^{DT}$ are first decremented by one for the entries that correspond to the current topic assignment
  - Then, a new topic is sampled from the current topic-distribution of a doc and the count matrices $\mathbf{C}^{WT}$ and $\mathbf{C}^{DT}$ are incremented with the new topic assignment.
- Each Gibbs sample consists the set of topic assignments to all *N* word tokens in the corpus, achieved by a single pass through all documents

Run Gibbs Sampling – An Example

Doc 1	Doc 2	Doc 3	Doc 4
Covid-19	Football	World	World
World	Football	World	Football
Scientist	Football	Outbreak	Covid-19
Covid-19	World	Covid-19	Football
Scientist	World	Covid-19	World

GOAL: Color each word with Blue, Purple and Green

1. Each document is a monochromatic as possible
2. Each word is a monochromatic as possible

Doc 1	Doc 2	Doc 3	Doc 4
Covid-19	Football	World	World
World	Football	World	Football
Scientist	Football	Outbreak	Covid-19
Covid-19	World	Covid-19	Football
Scientist	World	Covid-19	World

Uniformly random: assign each word in each document a topic

Covid-19	World	Scientist	Football	Football
Covid-19	World	Scientist	Football	
Covid-19	World		Football	
Covid-19	World		Football	
Covid-19	World		Football	
	World			
	World			

Doc 1	Doc 2	Doc 3	Doc 4
Covid-19	Football	World	World
World	Football	World	Football
Scientist	Football	Outbreak	Covid-19
Covid-19	World	Covid-19	Football
Scientist	World	Covid-19	World

Topic 1

How much is Topic 1 in Doc 1?

$$4 + \alpha$$

5= topics in Doc 1

How much is **"Covidd-19"** in Topic 1?

$$2 + \beta$$

7= words in topic 1

Product: 4/5 * 2/7 = 8/35

Topic 2

How much is Topic 2 in Doc 1?

$$0 + \alpha$$

5= topics in Doc 1

How much is **"Covidd-19"** in Topic 2?

$$1 + \beta$$

6= words in topic 2

Product: 0/5 * 1/6 = 0

Topic 3

How much is Topic 3 in Doc 1?

$$1 + \alpha$$

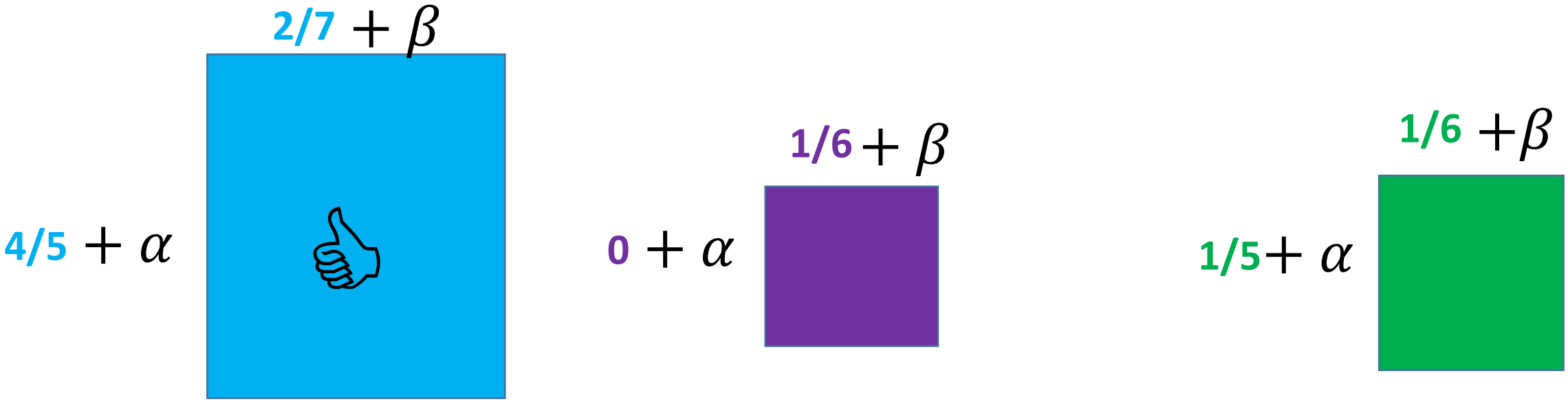
5= topics in Doc 1

How much is **"Covidd-19"** in Topic 3?

$$1 + \beta$$

6= words in topic 3

Product: 1/5 + 1/6 = 1/30



Topic 1

How much is Topic 1 in Doc 1?

$4 + \alpha$

5= topics in Doc 1

How much is **"Covidd-19"** in Topic 1?

$2 + \beta$

7= words in topic 1

Topic 2

How much is Topic 2 in Doc 1?

$0 + \alpha$

5= topics in Doc 1

How much is **"Covidd-19"** in Topic 2?

$1 + \beta$

6= words in topic 2

Topic 3

How much is Topic 3 in Doc 1?

$1 + \alpha$

5= topics in Doc 1

How much is **"Covidd-19"** in Topic 3?

$1 + \beta$

6= words in topic 3

Doc 1	Doc 2	Doc 3	Doc 4
Covid-19	Football	World	World
World	Football	World	Football
Scientist	Football	Outbreak	Covid-19
Covid-19	World	Covid-19	Football
Scientist	World	Covid-19	World

A moment later

Doc 1	Doc 2	Doc 3	Doc 4
Covid-19	Football	World	World
World	Football	World	Football
Scientist	Football	Outbreak	Covid-19
Covid-19	World	Covid-19	Football
Scientist	World	Covid-19	World

Covid-19	World	Scientist	Football	Outbreak
Covid-19	World	Scientist	Football	
Covid-19	World		Football	
Covid-19	World		Football	
Covid-19	World		Football	
	World			
	World			

Doc 1

Covid-19
World
Scientist
Covid-19
Scientist

80% Topic 1
20% Topic 3

Doc 2

Football
Football
Football
World
World

80% Topic 2
20% Topic 3

Doc 3

World
World
Outbreak
Covid-19
Covid-19

80% Topic 3
20% Topic 1

Doc 4

World
Football
Covid-19
Football
World

60% Topic 2
20% Topic 1
20% Topic 3

Science

Topic 1

Covid-19 (2)
World (2)
Scientist (2)

Sports

Topic 2

Football (4)
World (3)

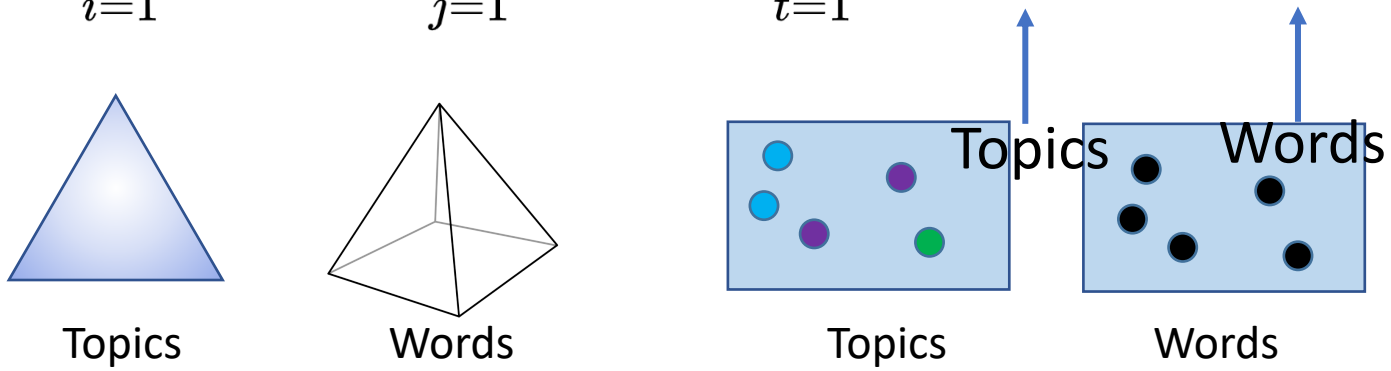
Health

Topic 3

Covid-19 (3)
Football (1)
World(2)
Outbreak (1)

Probability of a document

$$P(\boldsymbol{W}, \boldsymbol{Z}, \boldsymbol{\theta}, \boldsymbol{\varphi}; \alpha, \beta) = \prod_{i=1}^{K} P(\varphi_i; \beta) \prod_{j=1}^{M} P(\theta_j; \alpha) \prod_{t=1}^{N} P(Z_{j,t} \mid \theta_j) P(W_{j,t} \mid \varphi_{Z_{j,t}})$$



Gibbs sampling

Dirichlet distributions Multinomial distributions

Thank you!